

Fabrication of novel lithium lead bismuth borate glasses for nuclear radiation shielding

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ABSTRACT

In this research, glass samples composed of $(60-x)\text{B}_2\text{O}_3-x\text{Bi}_2\text{O}_3-30\text{PbO}-10\text{Li}_2\text{O}$ ($x = 0, 10, 20, 30, 40 \text{ mol\%}$) were prepared for utilize as shields for gamma rays and fast neutrons, using the melt-quenching technique. The studied samples were symbolized as Bi0.0–Bi10–Bi20–Bi30–Bi40 according to the Bi content values. The attenuation characteristics of gamma rays, such as the mass attenuation coefficients MAC and the linear attenuation coefficient LAC of the prepared glass samples, were obtained using Phy-X/PSD and XCOM software programs in the gamma energy range from 0.015 to 15 MeV. In addition, mean free path MFP, half-value layer HVL, effective atomic number Z_{eff} , and buildup factors were also investigated. The results indicated that the addition of Bi_2O_3 increased the MAC and Z_{eff} and reduced the HVL, MFP, and buildup factors. Also, the results showed that the prepared glass labeled Bi40 has the best shielding properties compared to the other samples. Compared with the concrete shielding such as barite and ordinary concrete, as well as, the RS-360 and RS-253 G18 commercial shielding glass, our prepared glass showed better gamma radiation protection properties. This indicates that our prepared samples can be used as alternative for other gamma ray shields.

1. Introduction

Protection from nuclear radiation as a result of nuclear reactor accidents or as shields in nuclear reactors or radiotherapy and x-ray rooms is very important to prevent the effect of ionizing radiation on the human body. The manufacture of high-efficiency materials to be used as a shield for nuclear radiation has become very important to avoid human damage from nuclear radiation (Al-Buriahi et al., 2020).

There are many materials whose shielding properties have been studied such as alloys, steel, polymers, stones and glass (Saleh et al., 2021). Recently, interest has focused on studying glass doped with heavy elements and using it as gamma ray shields because it is characterized by transparency in the visible region. When doped glass with oxides of heavy metals such as lead and bismuth oxides, the glass exhibits a high atomic number and high density due to the high molecular mass of the heavy metals. These features make glass an effective material for absorbing harmful gamma rays (Algradee et al., 2020; Saleh

et al., 2021).

The borate glass is a basis for most commercially available glass, which is one of the most important glass former. Borate glass possesses unique properties such as low melting point, high thermal stability, and the possibility of adding heavy elements as a modifier. The addition of heavy metal oxides such as lead and bismuth in glass improves glass density to be used as shields for gamma rays. Also, add some oxides such as lithium and sodium oxide to reduce the melting point for the glass prepared (Sathiyapriya et al., 2021).

Recently, several recent types of researches have focused on the use of glass doped with oxides of heavy elements such as lead, bismuth, and barium for use as gamma ray shields. Saleh et al., 2021 studied LNZN glasses doped with $(\text{CdO} + \text{Te})$ for their gamma-ray shielding properties, the results showed that sample LNZN5 has the best result for protection from gamma radiation. Al-Buriahi et al., 2020 prepared lead sodium borate glasses doped with chromium oxide, the results revealed that the prepared glass could be useful for shielding applications against nuclear

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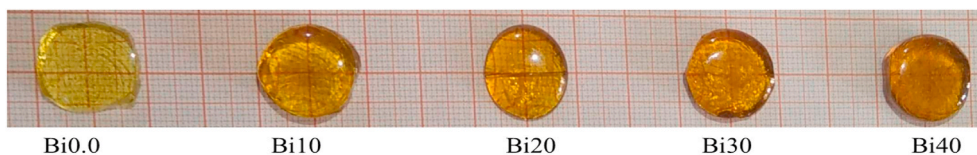


Fig. 1. Glass system image.

Table 1

Chemical compositions and density for glass samples.

Sample code	Mol%				Wt%					$\rho(\text{g}/\text{cm}^3)$
	B_2O_3	Bi_2O_3	PbO	Li_2O	B	O	Pb	Bi	Li	
Bi0.0	60	0.0	30	10	0.1161	0.3151	0.5564	0.0	0.0124	4.3831
Bi10	50	10	30	10	0.0714	0.2326	0.4107	0.2761	0.0092	4.9247
Bi20	40	20	30	10	0.0453	0.1843	0.3255	0.4377	0.0073	5.3408
Bi30	30	30	30	10	0.0281	0.1526	0.2695	0.5437	0.0060	5.4396
Bi40	20	40	30	10	0.0160	0.1302	0.2300	0.6186	0.0051	5.9430

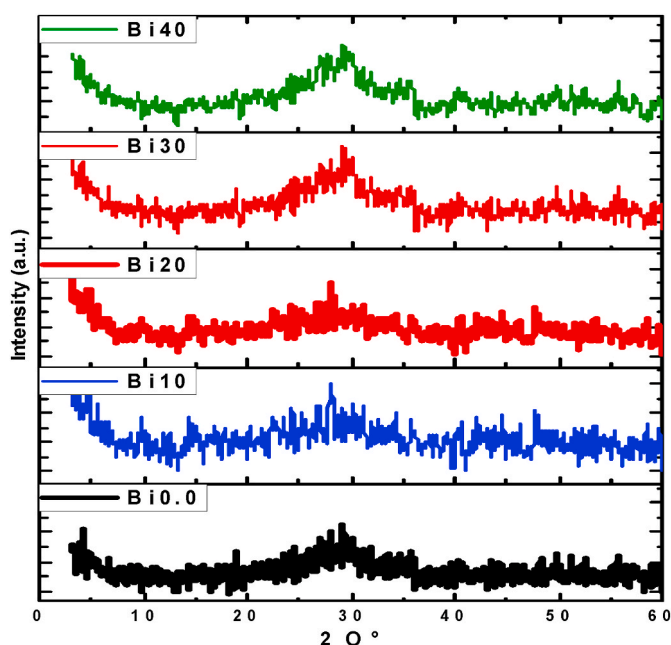


Fig. 2. XRD patterns for the prepared samples.

radiations. Sathiyapriya et al. (2021) synthesized the strontium barium borate (BNBD) glasses, they concluded that BNBD glass with a high concentration of BaCO_3 could be considered as an alternative material in nuclear radiation protection applications. Kaur et al. (2019) prepared Bi_2O_3 – B_2O_3 glasses doped with BaO and they concluded that the prepared glass system is the best absorber of gamma rays and can be used in gamma ray shielding applications. Lakshminarayana et al. (2020) studied five B_2O_3 – Bi_2O_3 glasses for photon and charged particle shielding, the results indicated that the glass $57.5\text{Bi}_2\text{O}_3$ – $42.5\text{B}_2\text{O}_3$ (mol %) has a superior attenuation for all kinds of radiations. Bagheri (2021) studies borate glasses doped with BaO , Bi_2O_3 , and PbO , the results showed that the atomic cross section and effective atomic number of the glass samples are increased by increasing the contents of BaO , PbO and Bi_2O_3 . Sayyed et al. (2018) investigated the BaO/SrO – Bi_2O_3 – B_2O_3 glasses for gamma and neutron radiation shielding purposes, they found that the addition of Bi_2O_3 leads to an increase in the effective atomic number values in both BaO/SrO – Bi_2O_3 – B_2O_3 glass systems. Algradee et al. (2020) synthesis and studied optical and gamma-ray shielding features of Nd^{3+} doped lithium-zinc-borophosphate glasses by using Phy-X/PSD software program. The results indicated the values of

nuclear shielding parameters improved with the increase the Nd_2O_3 content.

In the current work, lithium, lead, and bismuth oxides were added to the borate glass former and prepared by the melt-quenching method. The properties of nuclear shielding for prepared samples have been discussed by investigating the mass attenuation coefficient MAC, a half-value layer HVL, linear attenuation coefficient LAC, the mean free path MFP as well as an effective atomic number Z_{eff} , energy absorption, and exposure buildup factors (EABF and EBF) using XCOM and Phy-X/PSD software programs for energy photon from 0.015 to 15 MeV. Additionally, the structural properties of the prepared samples were studied using X-ray diffraction (XRD) (Şakar et al., 2020).

2. Materials and methods

2.1. Samples preparation

Lithium–Lead–Bismuth–Borate glass composed of $10\text{Li}_2\text{O}$ – 30PbO – $x\text{Bi}_2\text{O}_3$ – $(60-x)\text{B}_2\text{O}_3$, $x = 0.0, 10, 20, 30$ and 40 mol%, were fabricated by the melt-quenching method. The high purity of Li_2CO_3 , PbO , Bi_2O_3 , and H_3BO_2 were used to syntheses the prepared samples. After mixing and grinding materials, mixed quantities were placed in porcelain crucibles and inserted into an electric furnace for melting and held at 1050°C for 1 h. Samples are periodically swirling every 10 min to ensure their homogeneities. The samples have been poured on a stainless steel plate, and then were immediately transferred to an annealing furnace held at 350°C , for 1 h, then the furnace was switched off to allow the glass to cool gradually to room temperature. The glass samples were kept in the dried furnace for one day to prevent moisture entry into the samples. The fabricated samples were labeled as Bi0.0, Bi10, Bi20, Bi30, Bi40 According to the value of x in the glass formation. Fig. 1 shows the images of the prepared samples.

2.2. Characterization techniques

The XRD spectrum was obtained by a Bruker D8 Discover X-ray diffractometer. To explore the nature of the samples, the spectrum was found using 2θ regions of 3 – 60° . All measurements were made using X-rays with a wavelength of $1.54 \text{ \AA}$ of $\text{Cu K}\alpha$.

The density of the prepared samples was measured using Archimedes' principle. The samples were measured firstly at room temperature (25°C) in the air and then in the buoyant liquid (toluene) using an electronic balance with a sensitivity of (0.0001). The density of the prepared samples was calculated using the following equation (Algradee et al., 2020):

Table 2

Mass attenuation coefficients from Phy-X_PSD and XCOM programs.

Energy(MeV)	Bi0.0		Bi10		Bi20		Bi30		Bi40	
	Phy-X	XCOM	Phy-X	XCOM	Phy-X	XCOM	Phy-X	XCOM	Phy-X	XCOM
0.015	62.716	62.720	78.308	78.310	87.429	87.430	93.415	93.410	97.645	97.630
0.02	48.368	48.370	60.416	60.420	67.464	67.470	72.090	72.080	75.359	75.350
0.03	17.013	17.010	21.258	21.260	23.741	23.740	25.371	25.370	26.523	26.520
0.04	8.094	8.094	10.101	10.100	11.275	11.280	12.045	12.040	12.590	12.590
0.05	4.563	4.563	5.679	5.679	6.332	6.332	6.761	6.760	7.064	7.063
0.06	2.873	2.873	3.564	3.564	3.967	3.968	4.233	4.232	4.420	4.419
0.08	1.418	1.418	1.741	1.741	1.930	1.930	2.054	2.054	2.141	2.141
0.1	3.154	3.154	3.911	3.911	4.353	4.353	4.643	4.643	4.849	4.848
0.15	1.180	1.180	1.444	1.444	1.599	1.599	1.700	1.700	1.772	1.772
0.2	0.609	0.609	0.733	0.733	0.806	0.806	0.854	0.854	0.887	0.887
0.3	0.271	0.271	0.313	0.313	0.338	0.338	0.355	0.355	0.366	0.366
0.4	0.171	0.171	0.191	0.191	0.202	0.203	0.210	0.210	0.216	0.216
0.5	0.128	0.128	0.139	0.139	0.145	0.145	0.150	0.150	0.153	0.153
0.6	0.104	0.104	0.111	0.111	0.115	0.115	0.118	0.118	0.120	0.120
0.8	0.080	0.080	0.083	0.083	0.085	0.085	0.086	0.086	0.087	0.087
1	0.067	0.067	0.069	0.069	0.069	0.070	0.070	0.070	0.070	0.070
1.5	0.052	0.052	0.052	0.052	0.052	0.052	0.052	0.052	0.052	0.052
2	0.045	0.045	0.045	0.045	0.046	0.046	0.046	0.046	0.046	0.046
3	0.039	0.039	0.040	0.040	0.041	0.041	0.041	0.041	0.042	0.042
4	0.037	0.037	0.038	0.038	0.039	0.039	0.040	0.040	0.041	0.041
5	0.036	0.036	0.038	0.038	0.039	0.039	0.040	0.040	0.041	0.041
6	0.035	0.035	0.038	0.038	0.040	0.040	0.041	0.041	0.041	0.041
8	0.036	0.036	0.039	0.039	0.041	0.041	0.042	0.042	0.043	0.043
10	0.036	0.036	0.041	0.041	0.043	0.043	0.044	0.045	0.046	0.046
15	0.039	0.039	0.045	0.045	0.048	0.048	0.050	0.050	0.051	0.051

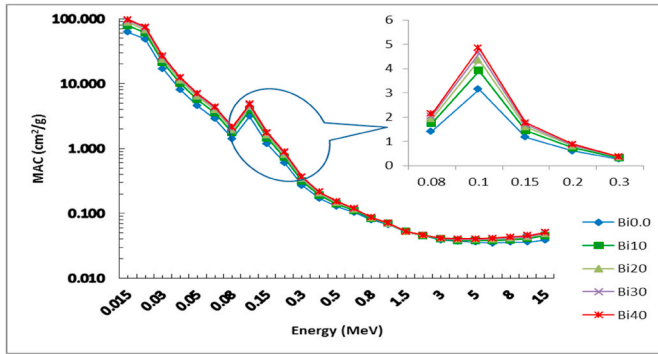


Fig. 3. MAC versus photon energy for all glass samples.

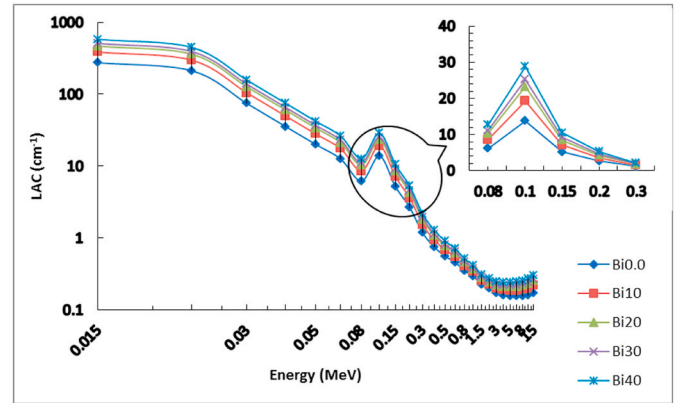


Fig. 5. LAC versus photon energy for all glass samples.

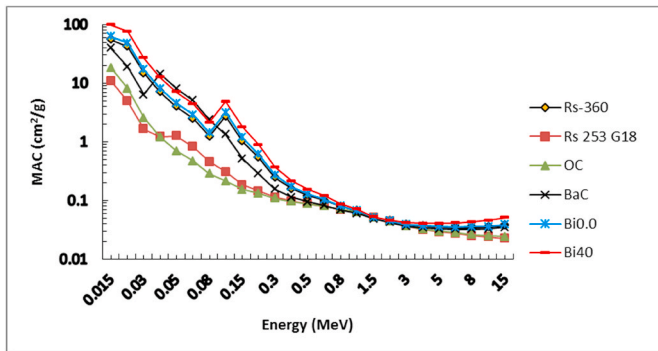


Fig. 4. Comparison of the minimum and maximum values of the MAC of the prepared samples and commercial materials.

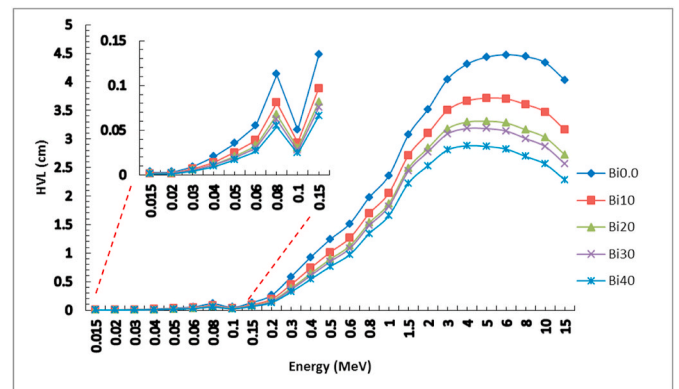


Fig. 6. HVL versus Photon energy for all glass samples.

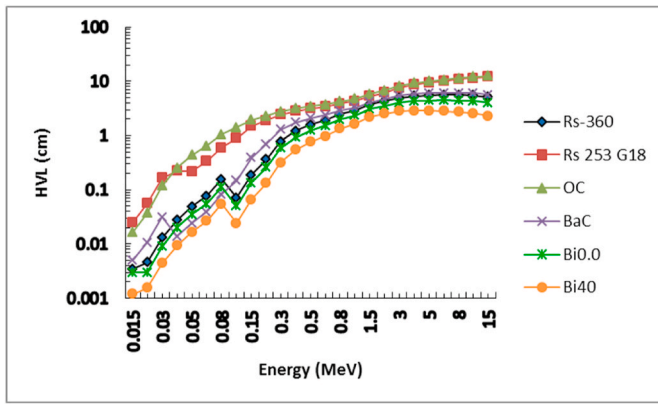


Fig. 7. Comparison of the minimum and maximum values of the HVL of the prepared samples and commercial materials.

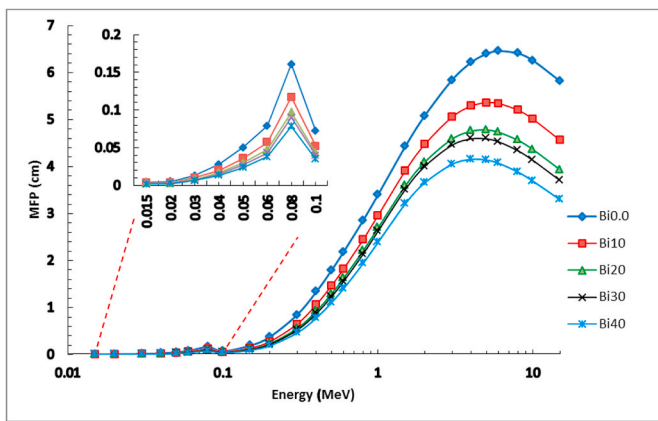


Fig. 8. MFP versus photon energy for all glass samples.

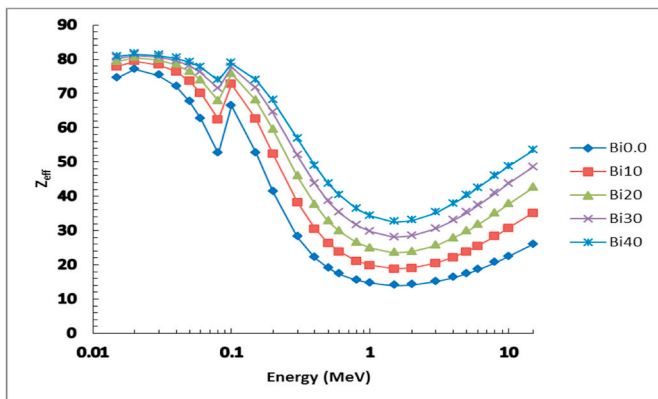


Fig. 9. Z_{eff} versus photon energy for all glass samples.

$$\rho = \frac{A}{A-B} \times \rho_0 \quad (1)$$

where A is the weight of sample in air, B is the weight of sample in toluene, ρ_0 is the density of toluene (0.87 g/cm^3).

2.3. Computational work

A user friendly online Photon Shielding and Dosimetry (PSD) software available at <https://phy-x.net/PSD> has been utilized for calculation of nuclear radiation shielding parameters (Şakar et al., 2020) as

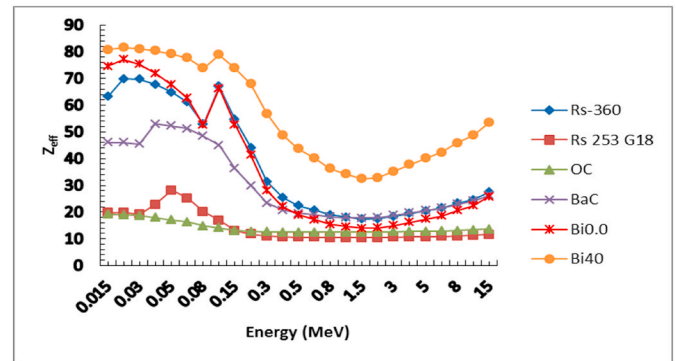


Fig. 10. Comparison of the minimum and maximum values of the Z_{eff} of the prepared samples and commercial materials.

well as XCOM online program (Berger and Hubbell, 1987). These parameters will be discussed in details as follows.

The linear attenuation coefficient LAC (μ) defines the percentage of X or gamma rays that are absorbed or scattered per unit thickness of the absorber. It can be represented by the following equation (Al-Buriah and Tonguc, 2020):

$$\mu = \frac{\ln \frac{I_0}{I}}{x} \quad (2)$$

where $\mu \text{ (cm}^{-1}\text{)}$ is the LAC of materials and I_0 and I are the initial and final intensity, respectively and x is the thickness of the material.

There is another parameter called the mass attenuation coefficient MAC, which is one of the most important factors that describe the ability to absorb and scatter gamma rays within a material. The mass attenuation coefficient (cm^2/g) for any materials is obtained by dividing the linear attenuation coefficient by the density of the substance according to the following equation (Singh et al., 2021):

$$MAC = \frac{\mu}{\rho} \quad (3)$$

where ρ is the material density (g/cm^3).

The half value layer (HVL) of any given material is defined as the thickness of the material that attenuates 50% of incident radiations. HVL is expressed in units of distance (cm), and given by the following relation (Al-Buriah et al., 2020):

$$HVL \text{ (cm)} = \frac{0.693}{\mu} \quad (4)$$

The mean free path (MFP) is the distance traveled by a moving particle or photon between two successive collisions with other particles, and it can be given by the following relation:

$$MFP \text{ (cm)} = \frac{1}{\mu} \quad (5)$$

The effective atomic number Z_{eff} of a substance has a strong and fundamental relationship with the nature of radiation interactions within this medium. For any compound substance consisting of more than one element, the Z_{eff} must be specified for that substance. The effective atomic number Z_{eff} of the compound is given by the following equation (Boukhris et al., 2020):

$$Z_{\text{eff}} = \frac{\sum f_i A_i \left(\frac{\mu}{\rho}\right)_i}{\sum \frac{f_i}{Z_i} \left(\frac{\mu}{\rho}\right)_i} \quad (6)$$

where f_i is a molar fraction, A_i is atomic mass and Z_i is atomic number.

Energy absorption buildup factor (EABF) and exposure buildup factor (EBF) values have been shown to be useful in selecting an alternative

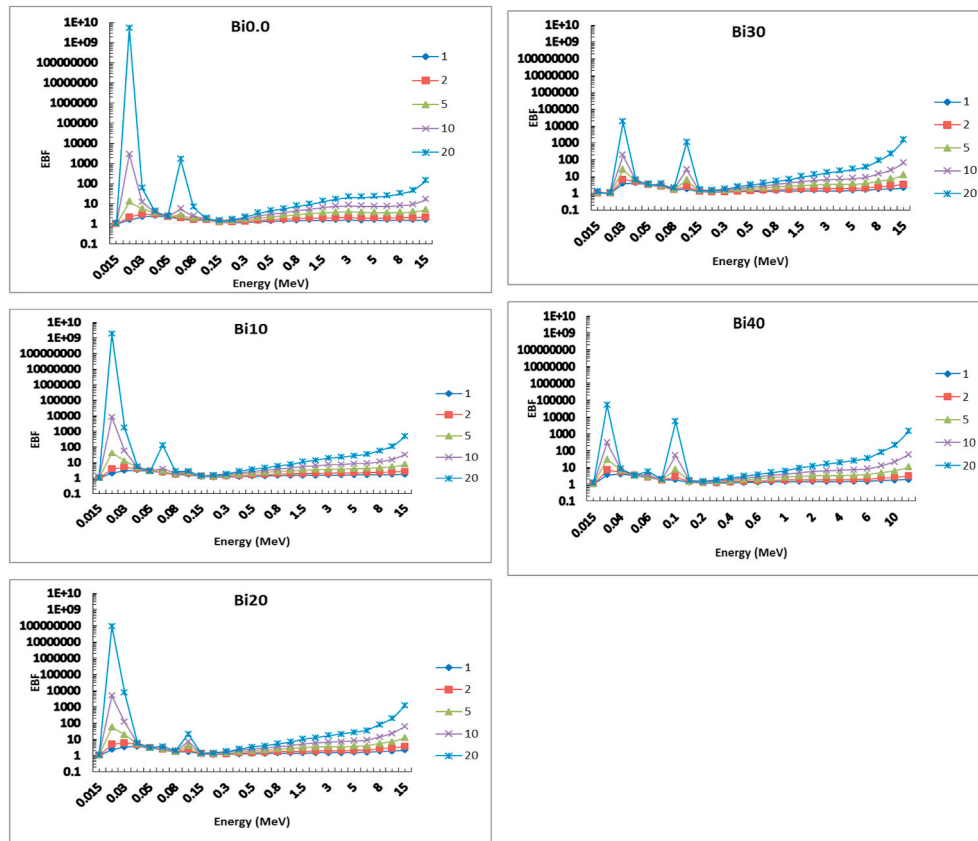


Fig. 11. Exposure build-up factor (EBF) versus photon energy for all glass samples.

composite material rather than an element at the required energy. The EBF and EABF values for a protective layer of gamma rays, with a time of up to 40 MFP and energy of 0.015–15 MeV, are calculated using the geometric progression (G-P) fitting parameters (b , c , a , x_k and d) according to the following equations (AlMisned et al., 2021):

$$B(E, X) = 1 + \frac{(b-1)(k^x - 1)}{k-1} \text{ for } k \neq 1 \quad (7)$$

$$k(E, X) = cx^a + d \frac{\tanh\left(\frac{x}{x_k} - 2\right) - \tanh(-2)}{1 - \tanh(-2)} \quad (8)$$

$$B(E, X) = 1 + (b-1)x \text{ for } k = 1 \quad (9)$$

where x is the distance in the medium in terms of MFP, K is the dose multiplicative factor, and b , c , a , x_k , and d are G-P fitting parameters.

Effective removal of neutron cross-section Σ_R :

Effective removal of neutron cross-section refers to the process of removing neutrons from materials at the appropriate time. The Σ_R values are a measure of the interaction probability between the fast neutrons and the target. The Σ_R values for a substance can be calculated using the following equation (AlMisned et al., 2021):

$$\Delta_R = \Delta w_i \left(\Delta_{R/\rho} \right) \quad (10)$$

where w_i is partial density (g/cm³) of the i th element and $\Sigma_{R/\rho}$ is mass removal cross-section (cm²/g).

3. Results and discussion

In this research, glass samples were prepared based on (60-x)B₂O₃-xBi₂O₃-30PbO-10Li₂O ($x = 0, 10, 20, 30, 40$ mol%) and examined for

their utilize as gamma ray shields. Five different glass samples labeled Bi0.0, Bi10, Bi20, Bi30, and Bi40 were prepared. The material density and chemical composition ratios of the oxides for the above-mentioned glass are given in Table 1. It is clear from Table 1 the density of the prepared samples ranged between 4.3831 and 5.9430 g/cm³ and increased with increasing bismuth oxide content. Due to the replacement of B₂O₃ with Bi₂O₃, the glass system becomes denser.

Fig. 2 shows the XRD patterns of the prepared samples. The XRD patterns of all samples reveal the absence of any sharp peaks, which explains the amorphous nature of the prepared glass.

Table 2 shows the mass attenuation coefficients (MAC) for glass samples prepared using the XCOM program and compared them with the results obtained using the Phy-X/PSD software program in the energy range from 0.015 to 15 MeV. From the table, we observed that the MAC values determined using Phy-X/PSD and XCOM were very identical. However, there were very slight differences in some energy values between the two methods used. Phy-X/PSD program is adopted in all of our calculations for nuclear shielding parameters. The results of MAC values are presented in Fig. 3. From Table 2 and Fig. 3, it is clear that the values of the MAC ranged between 62.716 and 0.039 cm²/g, 78.308–0.045 cm²/g, 87.429–0.048 cm²/g, 93.415–0.050 cm²/g and 97.645–0.051 cm²/g, for Bi0.0, Bi10, Bi20, Bi30 and Bi40 glass samples respectively. The highest values of the MAC are for the Bi40 sample with the highest density and the lowest MAC values are for the Bi0.0 sample with the lowest density. From Fig. 3, we observed that the highest value of the MAC was at the energy value 0.015 MeV, then it drops rapidly until it reaches the value 0.08 MeV, then increases suddenly before it reaches 0.1 MeV. This rapid decrease is attributed to the photoelectric absorption effect (PEA) in the low-energy zone and the change in the interaction cross-section is proportional to $1/E^{3.5}$ in that region. The sudden increase around 90 keV is attributed to the k-absorption edges of Pb and Bi, (88 and 90.52 keV). After that, the MAC values slowly

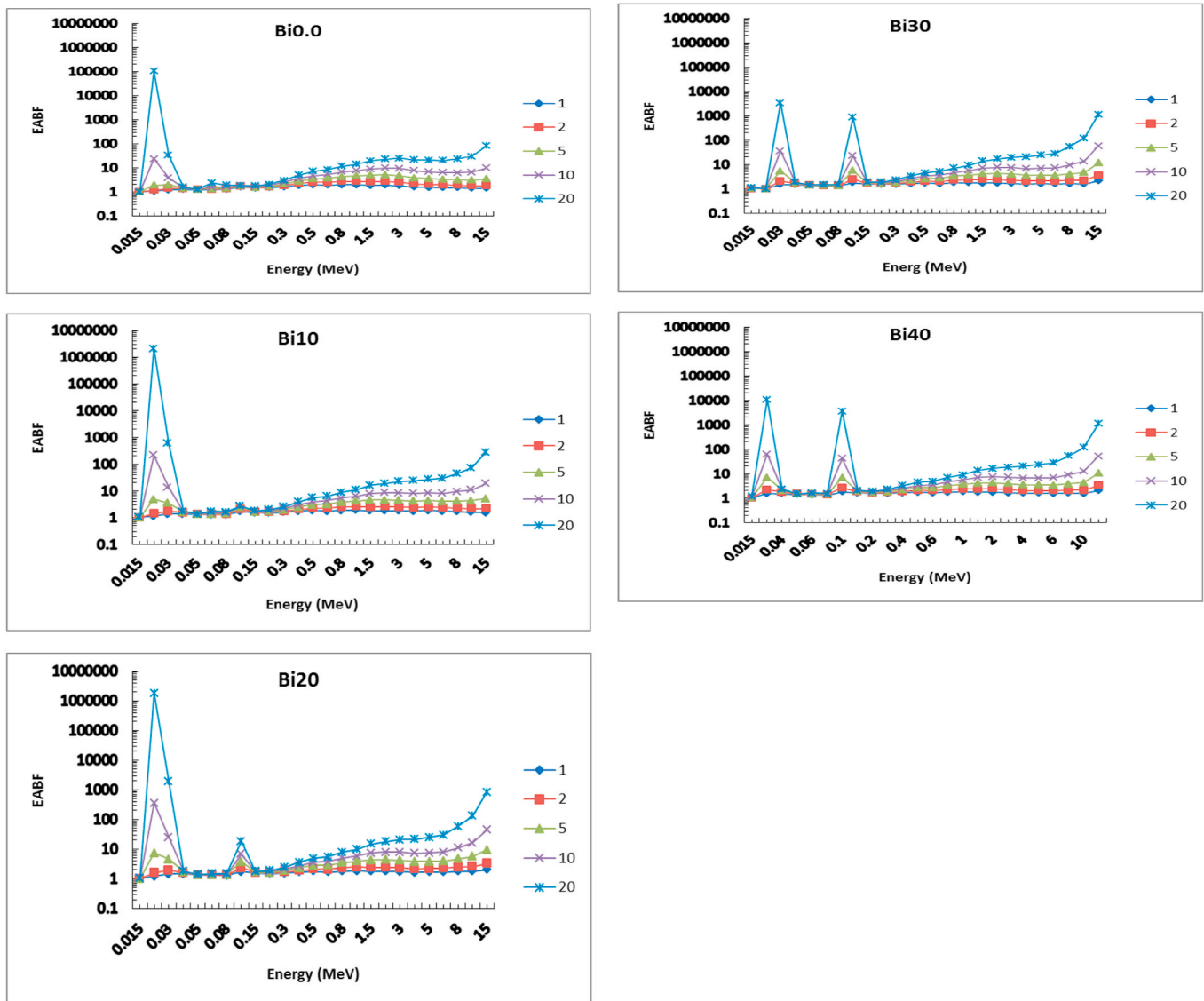


Fig. 12. Energy absorption build-up factor (EABF) versus photon energy for all glass samples.

decrease at medium energies where the Compton scattering (CS) process is dominant. After energy values of 1.022 MeV, where pair production (PP) occurs, we observed a slow increase in MAC values with increasing energy. In this region, the MAC values are depending on Z^2 .

Comparing the minimum and maximum values of the MAC of the prepared samples (Bi0.0 and Bi40) with the MAC of ordinary concrete, barite, as well as commercial glasses such as RS-360 and RS-253 G18, as in Fig. 4, we noticed that the values for the present samples are much better than the aforementioned commercial samples except values for the sample Bi0.0 at energy from 0.04 to 0.08 MeV.

Fig. 5 depicts the values of linear attenuation coefficients LAC (cm^{-1}) for the energy range from 0.015 to 15 MeV. From the figure, the highest value of LAC was 580.31 cm^{-1} for the Bi40 sample and the lowest value was 274.89 cm^{-1} for the Bi0.0 sample at the energy 0.015 MeV. The behavior of the LAC is similar to the MAC at all energy gamma rays values.

Fig. 6 shows the half-value layers HVL of the prepared glass samples in the energy range from 0.015 to 15 MeV. The lowest values of the HVL were at energy 0.015, which is close to zero for all glass samples. After that, it starts to slowly increase up to the energy value of 0.15 MeV, and then the values of the HVL increase rapidly due to the increase in the value of the photon energy up to 5 MeV, after this value it starts

decreasing again with an increase in the photon energy due to the Compton scattering of gamma rays, as well as the pair production. The lowest HVL values for Bi40 and Bi0.0 glass samples were obtained as 0.0011 and 0.0030 cm, respectively, at the energy value of 0.015 MeV. As for 0.1 MeV energy, the values were 0.024 and 0.05 cm for samples Bi40 and Bi0.0, respectively. These results are quite satisfactory as a good protective material in medical diagnostic rooms or treatment with gamma and x-rays. Fig. 7 shows the comparison of the minimum and maximum values of the HVL of the prepared samples with the HVL of the ordinary concrete, barite, as well as commercial glass such as RS-360 and RS-253 G18. From the figure, we notice that the values of the present samples are much better than the commercial samples mentioned above, except the values of the sample Bi0.0 at the energy ranges from 0.04 to 0.08 MeV.

The Mean Free Path MFP is defined as the distance traveled between two successive collisions of gamma photons. The smaller the MFP values mean a better attenuating gamma photons. As in Fig. 8, the values of the MFP increase slowly with the increase in the photon energy until reaches the energy value 0.1 MeV, then it increases rapidly with the increase in energy up to 5 MeV and then decreases again due to the scattering of the photon. This behavior is due to different photon interaction processes with the absorbent in different energy regions. The lowest mean free

Table 3

(EBF and EABF) G-P fitting parameters (a,b, c, Xk and d) of Bi0.0 glass sample.

Energy (MeV)	Z_{eq}	G-P Fitting Parameters for EBF					G-P Fitting Parameters for EABF				
		a	b	c	d	Xk	a	b	c	d	Xk
0.015	27.35	-0.441	1.003	1.784	0.306	8.163	-0.441	1.003	1.784	0.306	8.163
0.02	33.01	0.505	1.550	0.776	-0.694	11.662	0.294	1.083	0.865	-0.334	14.545
0.03	34.02	0.164	2.150	0.675	-0.123	20.683	0.186	1.231	0.685	-0.152	18.482
0.04	34.77	0.165	2.462	0.326	-0.065	18.818	0.153	1.259	0.357	-0.169	24.358
0.05	35.30	-0.054	2.153	0.186	-0.036	12.396	0.032	1.237	0.193	-0.001	10.453
0.06	35.75	0.726	1.843	0.161	-0.142	15.788	0.560	1.237	0.161	-0.173	14.745
0.08	36.48	0.548	1.481	0.190	-0.200	14.236	0.437	1.279	0.201	-0.188	14.173
0.1	61.78	0.197	1.572	0.179	-0.008	17.846	0.190	1.585	0.178	-0.012	17.117
0.15	63.59	0.467	1.198	0.154	-0.257	13.825	0.671	1.465	0.063	-0.311	14.005
0.2	64.55	0.236	1.159	0.381	-0.126	14.151	0.495	1.487	0.140	-0.269	13.771
0.3	65.68	0.147	1.222	0.534	-0.069	13.776	0.330	1.640	0.272	-0.186	13.368
0.4	66.31	0.105	1.300	0.652	-0.056	14.208	0.253	1.830	0.390	-0.163	13.763
0.5	66.72	0.082	1.365	0.723	-0.047	14.144	0.201	1.956	0.484	-0.134	13.818
0.6	67.00	0.065	1.409	0.775	-0.038	13.731	0.146	1.894	0.590	-0.098	13.635
0.8	67.30	0.045	1.470	0.846	-0.030	13.697	0.110	1.983	0.682	-0.079	13.593
1	67.42	0.030	1.494	0.902	-0.024	13.456	0.089	1.997	0.743	-0.070	13.517
1.5	66.17	0.005	1.478	1.012	-0.018	14.077	0.049	1.876	0.880	-0.053	13.719
2	62.18	0.001	1.510	1.042	-0.020	13.133	0.059	1.910	0.870	-0.069	13.358
3	53.84	0.012	1.534	1.030	-0.040	13.216	0.072	1.824	0.856	-0.093	13.394
4	48.99	0.016	1.474	1.034	-0.045	13.422	0.062	1.587	0.893	-0.085	13.626
5	46.32	0.046	1.503	0.950	-0.072	13.646	0.088	1.563	0.829	-0.108	13.842
6	44.81	0.058	1.490	0.925	-0.081	13.857	0.100	1.514	0.805	-0.118	14.037
8	43.02	0.079	1.530	0.883	-0.098	14.151	0.111	1.480	0.794	-0.125	14.297
10	42.09	0.059	1.502	0.966	-0.078	14.210	0.087	1.413	0.878	-0.101	14.349
15	41.31	0.036	1.556	1.111	-0.060	14.174	0.063	1.394	1.015	-0.084	14.304

Table 4

(EBF and EABF) G-P fitting parameters (a,b, c, Xk and d) of Bi10 glass sample.

Energy (MeV)	Z_{eq}	G-P Fitting Parameters for EBF					G-P Fitting Parameters for EABF				
		a	b	c	d	Xk	a	b	c	d	Xk
0.015	30.64	-0.231	1.002	1.795	0.198	11.368	-0.234	1.002	1.797	0.202	11.626
0.02	37.16	0.268	2.045	1.391	-0.359	12.467	0.159	1.154	1.445	-0.174	16.313
0.03	37.93	0.135	2.920	0.890	-0.172	26.592	0.145	1.377	0.899	-0.135	23.396
0.04	38.45	0.124	3.251	0.324	-0.047	21.246	0.128	1.382	0.337	-0.103	23.751
0.05	38.89	-0.201	2.685	0.103	0.015	12.153	-0.071	1.318	0.115	0.067	9.254
0.06	39.27	0.946	2.172	0.069	-0.146	16.797	0.699	1.290	0.081	-0.184	14.841
0.08	39.88	0.687	1.595	0.091	-0.236	14.340	0.523	1.292	0.125	-0.213	14.089
0.1	68.03	0.063	1.679	0.504	-0.022	17.879	0.067	1.705	0.489	-0.026	17.510
0.15	69.53	0.405	1.229	0.147	-0.160	14.566	0.470	1.539	0.087	-0.145	19.862
0.2	70.30	0.305	1.156	0.298	-0.170	13.857	0.580	1.508	0.099	-0.289	13.883
0.3	71.19	0.161	1.187	0.502	-0.077	13.671	0.375	1.585	0.225	-0.208	13.329
0.4	71.68	0.118	1.253	0.616	-0.062	14.160	0.282	1.722	0.345	-0.179	13.725
0.5	71.99	0.094	1.312	0.686	-0.052	14.134	0.226	1.836	0.436	-0.148	13.761
0.6	72.20	0.076	1.356	0.735	-0.042	13.700	0.161	1.759	0.550	-0.104	13.594
0.8	72.44	0.054	1.417	0.808	-0.033	13.707	0.124	1.867	0.639	-0.084	13.583
1	72.53	0.039	1.449	0.863	-0.028	13.365	0.102	1.899	0.703	-0.075	13.527
1.5	71.73	0.011	1.437	0.987	-0.020	14.238	0.056	1.810	0.852	-0.055	13.814
2	69.10	0.004	1.460	1.026	-0.020	13.282	0.063	1.821	0.853	-0.069	13.407
3	63.06	0.018	1.495	1.013	-0.044	13.277	0.093	1.801	0.801	-0.111	13.525
4	59.07	0.036	1.505	0.975	-0.060	13.740	0.109	1.750	0.774	-0.126	13.856
5	56.63	0.069	1.605	0.891	-0.088	13.947	0.142	1.840	0.706	-0.157	14.099
6	55.19	0.072	1.608	0.896	-0.090	14.123	0.136	1.747	0.731	-0.150	14.228
8	53.45	0.074	1.689	0.921	-0.094	14.170	0.121	1.696	0.793	-0.137	14.309
10	52.52	0.043	1.659	1.057	-0.066	14.135	0.086	1.594	0.920	-0.107	14.204
15	51.61	0.022	1.734	1.233	-0.052	13.854	0.057	1.552	1.099	-0.088	14.013

path was for the prepared glass sample Bi40, and the largest free path was for the sample Bi0.0.

The effective atomic number (Z_{eff}) is used to express the atomic number of compound materials, which shows the shielding capacity of the material. Fig. 9 displays the change of Z_{eff} values for the prepared glasses against the photon energy. The Z_{eff} values for the prepared samples ranged between 74.53 and 25.86, 77.88–35.14, 79.37–42.48, 80.22–48.44 and 80.76–53.37 for Bi0.0, Bi10, Bi20, Bi30 and Bi40 glass samples, respectively. The largest Z_{eff} values were at low energies 0.02 MeV, due to the absorption L edge of Pb (L: 15.86 keV) and Bi (L: 16.38 keV). After that, the Z_{eff} values decreased slowly until they reached the

energy value 0.08 MeV, then increased rapidly due to the k-absorption edge of Pb (K: 88.00 keV) and Bi (k: 90.52 keV). Between 0.1 and 1 MeV, the Z_{eff} values decrease rapidly, and after 1 MeV, the Z_{eff} increase again due to the pair production predominant which varies with Z^2 . Moreover, effective atomic number Z_{eff} of the prepared glass samples comparison with commercial materials used as shielding for gamma rays and the results presented in Fig. 10. From the figure, we observed that the values of the Z_{eff} of the present glass samples are superior to the values of the Z_{eff} of the aforementioned materials, except Z_{eff} values of Bi0.0 glass sample at the energy from 0.2 to 15 MeV.

The buildup factor is a correction multiplier caused by secondary

Table 5
(EBF and EABF) G-P fitting parameters (a,b, c, Xk and d) of Bi20 glass sample.

Energy (MeV)	Z_{eq}	G-P Fitting Parameters for EBF					G-P Fitting Parameters for EABF				
		a	b	c	d	Xk	a	b	c	d	Xk
0.015	32.73	-0.136	1.004	1.495	0.134	11.631	-0.143	1.004	1.500	0.142	12.200
0.02	39.93	0.124	2.345	1.764	-0.156	12.956	0.078	1.197	1.797	-0.077	17.384
0.03	40.26	0.120	3.343	1.008	-0.199	29.843	0.122	1.457	1.017	-0.126	26.100
0.04	40.62	0.102	3.679	0.323	-0.038	22.565	0.115	1.448	0.326	-0.067	23.421
0.05	40.96	-0.279	2.971	0.059	0.042	12.023	-0.127	1.362	0.072	0.104	8.612
0.06	41.26	1.061	2.346	0.021	-0.147	17.329	0.773	1.317	0.038	-0.189	14.892
0.08	41.79	0.759	1.655	0.038	-0.254	14.394	0.568	1.299	0.085	-0.227	14.045
0.1	71.61	0.122	1.731	0.726	-0.092	16.660	0.128	1.757	0.701	-0.095	16.797
0.15	72.83	0.366	1.247	0.145	-0.104	14.978	0.353	1.580	0.102	-0.051	23.135
0.2	73.47	0.341	1.155	0.255	-0.193	13.705	0.624	1.519	0.077	-0.300	13.941
0.3	74.18	0.169	1.169	0.485	-0.082	13.617	0.398	1.558	0.202	-0.219	13.309
0.4	74.58	0.124	1.230	0.598	-0.065	14.144	0.298	1.672	0.321	-0.187	13.702
0.5	74.83	0.099	1.286	0.668	-0.053	14.138	0.239	1.780	0.411	-0.155	13.732
0.6	74.99	0.081	1.329	0.717	-0.044	13.702	0.168	1.694	0.532	-0.106	13.576
0.8	75.18	0.058	1.391	0.791	-0.033	13.707	0.131	1.811	0.618	-0.086	13.572
1	75.25	0.043	1.425	0.846	-0.028	13.394	0.108	1.851	0.683	-0.077	13.527
1.5	74.67	0.014	1.416	0.974	-0.021	14.331	0.060	1.778	0.836	-0.057	13.862
2	72.74	0.006	1.436	1.016	-0.020	13.370	0.064	1.778	0.845	-0.069	13.432
3	68.18	0.016	1.452	1.018	-0.041	13.330	0.090	1.730	0.806	-0.108	13.534
4	64.88	0.032	1.456	0.992	-0.056	13.738	0.104	1.640	0.789	-0.121	13.868
5	62.79	0.064	1.546	0.911	-0.084	13.977	0.137	1.698	0.723	-0.152	14.143
6	61.50	0.071	1.593	0.904	-0.089	14.170	0.138	1.697	0.730	-0.152	14.290
8	59.94	0.070	1.794	0.945	-0.089	14.158	0.125	1.794	0.793	-0.141	14.280
10	59.09	0.034	1.879	1.106	-0.058	14.029	0.084	1.793	0.944	-0.107	14.085
15	58.20	0.007	2.148	1.324	-0.039	13.685	0.042	2.022	1.181	-0.077	13.754

Table 6
(EBF and EABF) G-P fitting parameters (a,b, c, Xk and d) of Bi30 glass sample.

Energy (MeV)	Z_{eq}	G-P Fitting Parameters for EBF					G-P Fitting Parameters for EABF				
		a	b	c	d	Xk	a	b	c	d	Xk
0.015	34.27	-0.070	1.005	1.288	0.089	11.813	-0.080	1.005	1.294	0.101	12.597
0.02	72.66	0.144	1.003	0.459	-0.082	23.749	0.144	1.003	0.459	-0.079	23.420
0.03	41.82	0.110	3.611	1.083	-0.216	31.901	0.108	1.508	1.091	-0.120	27.811
0.04	42.04	0.088	3.939	0.327	-0.033	23.385	0.107	1.490	0.325	-0.045	23.168
0.05	42.30	-0.292	3.125	0.057	0.044	12.159	-0.136	1.394	0.070	0.107	8.644
0.06	42.56	1.017	2.449	0.022	-0.141	16.566	0.732	1.345	0.041	-0.172	15.387
0.08	43.03	0.773	1.681	0.031	-0.242	14.546	0.595	1.313	0.074	-0.228	14.068
0.1	73.91	0.159	1.763	0.862	-0.135	15.909	0.167	1.789	0.831	-0.137	16.357
0.15	74.98	0.342	1.272	0.171	-0.077	15.371	0.311	1.611	0.132	-0.016	24.685
0.2	75.50	0.362	1.164	0.230	-0.195	13.766	0.603	1.534	0.075	-0.263	14.628
0.3	76.09	0.177	1.164	0.469	-0.087	13.585	0.418	1.551	0.187	-0.227	13.302
0.4	76.41	0.127	1.219	0.589	-0.067	14.156	0.311	1.652	0.306	-0.194	13.676
0.5	76.61	0.102	1.273	0.659	-0.054	14.154	0.249	1.753	0.396	-0.161	13.716
0.6	76.75	0.083	1.313	0.709	-0.044	13.722	0.171	1.658	0.522	-0.107	13.569
0.8	76.91	0.060	1.375	0.783	-0.033	13.702	0.136	1.778	0.606	-0.088	13.561
1	76.97	0.045	1.410	0.837	-0.028	13.466	0.112	1.822	0.670	-0.079	13.522
1.5	76.51	0.015	1.404	0.966	-0.021	14.414	0.065	1.763	0.823	-0.058	13.895
2	75.01	0.008	1.422	1.008	-0.020	13.415	0.068	1.759	0.834	-0.070	13.448
3	71.42	0.015	1.426	1.021	-0.040	13.381	0.088	1.690	0.810	-0.105	13.538
4	68.67	0.026	1.416	1.011	-0.051	13.730	0.102	1.598	0.794	-0.120	13.860
5	66.88	0.059	1.490	0.930	-0.080	13.968	0.132	1.624	0.736	-0.148	14.147
6	65.77	0.067	1.532	0.924	-0.085	14.194	0.134	1.604	0.743	-0.149	14.298
8	64.39	0.066	1.715	0.967	-0.086	14.131	0.119	1.641	0.814	-0.137	14.281
10	63.63	0.033	1.799	1.125	-0.059	13.965	0.078	1.601	0.974	-0.101	14.136
15	62.82	0.005	2.029	1.351	-0.037	13.705	0.015	2.149	1.289	-0.047	13.694

particles primarily associated with Compton scattering. The exposure buildup factor (EBF) and energy absorption buildup factor (EABF) are obtained by appropriate approximation of geometric progression (G-P) fitting parameters. EBF and EABF were determined for prepared glass samples using Phy-X/PSD software at photon energies from 0.015 to 15 MeV. Figs. 11 and 12 show the change in the EBF and EABF values of the prepared glass samples versus the photon energy and are calculated using G-P fitting parameters. Tables 3–7 have G-P fitting parameters for exposure buildup factor (EBF) and energy absorption buildup factor EABF. From the figures, due to the L-absorption edges of Pb and Bi elements, all samples show a very sharp peak at low energy 0.02 MeV.

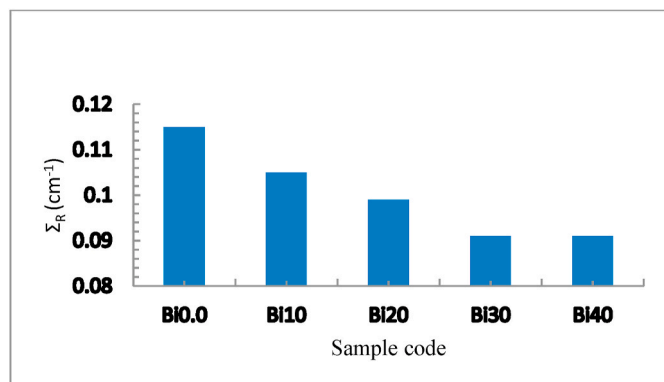
Another peak appeared at the energy of 0.09 MeV for all glass samples, due to the k-absorption edge of the Pb and Bi elements. From Figs. 11 and 12 we observed that the values of EBF and EABF for all glass samples at energies of 1 keV are very small due to the Z-proportional Compton scattering process. In the range from 1 to 15 MeV where the reaction of pair production increases and with increasing penetration depth, the photon buildup increases with secondary scattering. As a result, the EBF and EABF values for all glass samples begin to increase again. Also, from Figures, it was noticed that EBFs are higher than EABFs for all glass samples.

To assess the protective ability of prepared glass samples against

Table 7

(EBF and EABF) G-P fitting parameters (a, b, c, Xk and d) of Bi40 glass sample.

Energy (MeV)	Z_{eq}	G-P Fitting Parameters for EBF					G-P Fitting Parameters for EABF				
		a	b	c	d	Xk	a	b	c	d	Xk
0.015	35.46	-0.021	1.005	1.133	0.056	11.948	-0.033	1.005	1.141	0.070	12.893
0.02	74.43	—	—	—	—	—	—	—	—	—	—
0.03	42.93	0.100	3.548	1.190	-0.194	29.448	0.098	1.546	1.196	-0.108	25.874
0.04	43.07	0.089	3.888	0.452	-0.046	23.795	0.104	1.514	0.456	-0.042	21.793
0.05	43.27	-0.215	3.173	0.115	0.006	12.784	-0.080	1.427	0.127	0.061	9.727
0.06	43.48	0.878	2.515	0.050	-0.130	15.031	0.625	1.375	0.070	-0.142	16.154
0.08	43.89	0.778	1.698	0.029	-0.230	14.665	0.613	1.323	0.069	-0.228	14.090
0.1	75.58	0.140	1.798	0.997	-0.132	18.716	0.153	1.820	0.939	-0.125	15.627
0.15	76.48	0.326	1.301	0.211	-0.065	15.752	0.308	1.637	0.170	-0.012	25.328
0.2	76.93	0.377	1.172	0.214	-0.193	13.846	0.577	1.547	0.076	-0.228	15.261
0.3	77.41	0.183	1.160	0.458	-0.090	13.563	0.431	1.546	0.177	-0.233	13.297
0.4	77.68	0.130	1.212	0.583	-0.067	14.164	0.319	1.638	0.296	-0.199	13.658
0.5	77.85	0.103	1.263	0.653	-0.054	14.165	0.255	1.734	0.385	-0.164	13.705
0.6	77.95	0.085	1.303	0.703	-0.044	13.736	0.174	1.634	0.516	-0.108	13.565
0.8	78.07	0.061	1.364	0.777	-0.033	13.700	0.139	1.756	0.597	-0.089	13.554
1	78.14	0.047	1.399	0.831	-0.028	13.515	0.115	1.803	0.662	-0.080	13.519
1.5	77.77	0.016	1.396	0.960	-0.021	14.469	0.067	1.753	0.813	-0.060	13.918
2	76.59	0.009	1.413	1.001	-0.021	13.437	0.071	1.753	0.823	-0.072	13.460
3	73.67	0.014	1.409	1.023	-0.039	13.415	0.087	1.663	0.813	-0.103	13.540
4	71.33	0.022	1.389	1.024	-0.047	13.725	0.101	1.569	0.797	-0.119	13.855
5	69.81	0.055	1.449	0.944	-0.077	13.956	0.129	1.583	0.744	-0.145	14.144
6	68.85	0.062	1.480	0.939	-0.082	14.251	0.133	1.558	0.749	-0.148	14.295
8	67.64	0.065	1.627	0.975	-0.086	14.138	0.122	1.585	0.810	-0.141	14.291
10	66.98	0.032	1.693	1.135	-0.058	14.004	0.080	1.538	0.970	-0.104	14.162
15	66.24	0.008	1.903	1.353	-0.040	13.684	0.020	2.003	1.279	-0.053	13.730

**Fig. 13.** Effective removal cross-sections Σ_R for fast neutron for all prepared glass samples.

uncharged particles (fast neutrons) the macroscopic removal cross-section (Σ_R) of glass samples for fast neutrons were evaluated. Large values of Σ_R mean that the number of reactions per unit time increases and therefore the fast neutrons decrease. Fig. 13, shows the (Σ_R) values of the prepared glass samples. The Σ_R values for the prepared glass samples were 0.115 cm⁻¹, 0.105 cm⁻¹, 0.099 cm⁻¹, 0.091 cm⁻¹ and 0.091 cm⁻¹ for Bi0.0 – Bi40, respectively. Also, from the figure, we noticed the values of Σ_R were decrease with the increase in density and the addition of bismuth element. This is due to the replacement of boron by bismuth in glass samples, as boron has the greatest value of the Σ_R from bismuth. Glass labeled Bi0.0 appears to be more efficient at capturing neutrons than other glass samples. It should be noticed that the values of Σ_R for the sample Bi0.0 and Bi10 are better than Σ_R of water and paraffin (0.101 cm⁻¹) which are used as moderators of fast neutrons.

4. Conclusion

In this research, five glass samples composed of (60-x)B₂O₃-xBi₂O₃-30PbO-10Li₂O, where (x = 0, 10, 20, 30 and 40 mol%), were prepared for utilizing as nuclear radiation shields. The prepared samples

were examined using XCOM and Phy-X/PSD software programs. For this aim, the MAC values for the prepared glass samples in the energy range 0.015–15 MeV were obtained. The mass attenuation coefficient values for the studied samples ranged between 62.716 and 0.039 cm²/g, 78.308–0.045 cm²/g, 87.429–0.048 cm²/g, 93.415–0.050 cm²/g and 97.645–0.051 cm²/g for Bi0.0 – Bi40, respectively. Other nuclear radiation parameters such as LEC, HVL, MFP, Z_{eff} , EABF, and EBF, were also calculated. When all parameters were evaluated, it was revealed that the Bi40 glass system was more effective than Bi0.0 in gamma radiation shielding. The protection against fast neutrons was also studied by calculating the effective removal cross-section for fast neutrons and it was found that sample Bi0.0 was more effective for absorbing fast neutrons than sample Bi40. The results of the study were compared with commercial materials used as gamma shields, such as ordinary concrete and barite, as well as commercial glass such as RS-360 and RS-253 G18, and from the results we have obtained, we conclude that Bi40 glass show superior performance compared to other prepared glasses as well as commercial glasses in reducing photons.

Author contributions

Experimental work, E.E.S., and M. A. A.; Conceptualization, E.E.S., M. A. A., G. M. Y.; methodology, E. E. S., M. A. A., and S. A. E.; software, E.E.S.; validation, E.E.S., S. A. E., and M. A. A.; formal analysis, E.E.S. and M.A. A.; investigation, E. E. S. and M. A.A.; resources, G. M. Y., M. A. A., and E. E. S.; data curatain, E.E.S. and M.A.A; writing—original draft preparation, E.E.S.; writing—review and editing, S.A.E., and G. M. Y.; visualization, E. E. S.; supervision, S. A. E. and G. M. Y; All authors have read and agreed to the published version of the manuscript.

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Not applicable.

Data availability statement

The data presented in this study are available on request from the corresponding author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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